PRICAI 2026 Stroke Mortality · Deep Tabular Learning · FT-Transformer

DualTower-FT with an Adaptive Runtime Safeguard

A Deep Tabular Approach for ICU Stroke Mortality

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Code & clarifications
GitHub

0.6247 Best evaluated deep-model AUPRC

0% Crashes in tested safeguard-ON batches

Negative transfer No observed improvement from MIMIC-IV training

01 Problem & Motivation

Multi-center ICU EHR data are noisy — coding practices, equipment and populations differ across hospitals. Deep tabular models can fail or behave unreliably at **inference time** on inputs never seen in training:

- **Categorical out-of-range indices**
- **Numerical extremes**
- **NaN / ±∞ inputs**

02 Contributions

01 [CLS] aggregation
Learned [CLS] token replaces mean pooling in the numerical FT-Transformer tower.

02 Adaptive Runtime Safeguard
Preprocessing median-imputes missing values; a forward-pass backstop guards NaN/±∞, categorical out-of-range indices, and numerical extremes.

03 Cross-database evaluation
MIMIC-IV transfer study with a documented negative result.

03 Dataset & Setup

11,187 eICU stroke patients · **176** hospitals · **14.0%** mortality rate

Train 7,830 | Val 1,119 | Test 2,238

94 features = **3** categorical + **91** numerical.
Categorical: `gender, ethnicity, site_id`
Stratified split by ICU stay (70/10/20)

Transfer: MIMIC-IV — 7,125 patients, one academic medical center.

04 Methodology — DualTower-FT architecture

Figure 1 · method

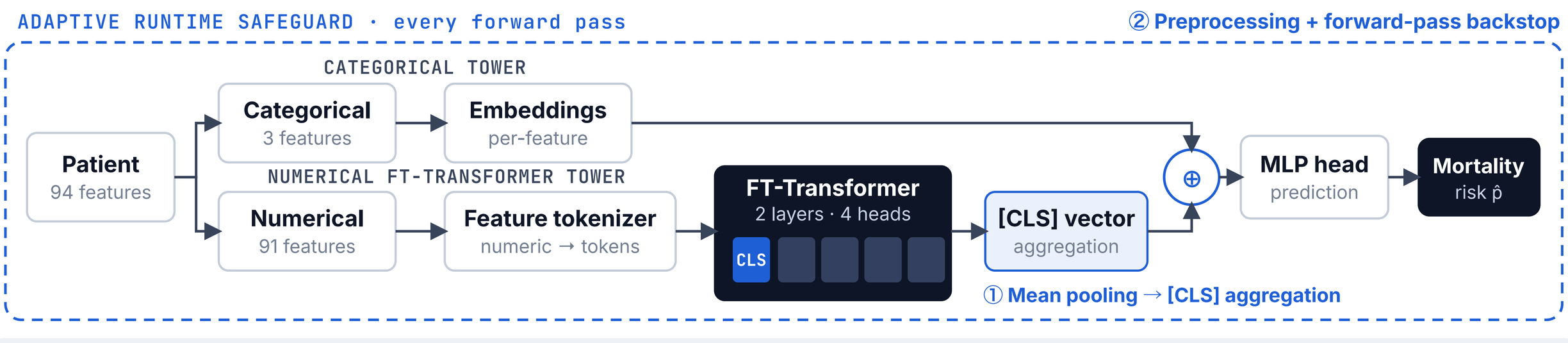
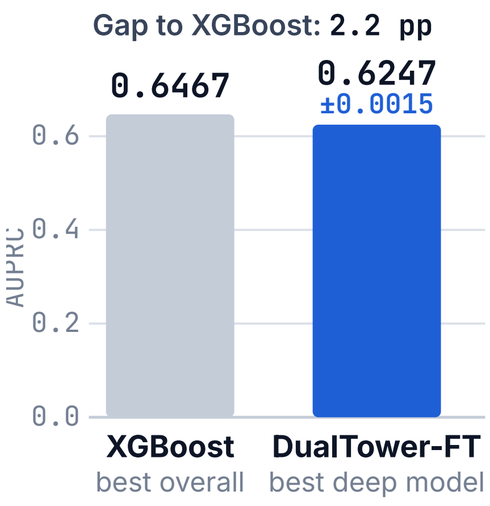


Figure 1. DualTower-FT architecture (author diagram). Contributions ① and ② are highlighted in blue. Implementation note: the saved scaler standardises categorical indices; runtime clamping reduces the 176 hospital site IDs to two effective indices. See the GitHub README.

05 Main Performance

Figure 2 · eICU test cohort



0.6247 ±0.0015
AUPRC · primary metric (14.0% prevalence)

0.5417 ±0.0156
F1 · Youden's J on validation, per seed (0.37–0.76)

0.8848
AUROC

Bootstrap 95% CI for AUPRC: 0.5764–0.6735. XGBoost (0.6467) lies within this bootstrap CI. *This is not a paired significance test.*

Calibration imperfect (Brier 0.148, ECE 0.195); recalibration needed before deployment.

Figure 2. eICU test performance (N = 2,238). *Baseline results are taken from our prior dual-tower work (same cohort, 94 features) and were not retrained; the cross-model comparison is therefore not fully controlled.*

06 Ablation

Table 1 · architecture ablation

Variant	Change	AUPRC	Δ pp
DualTower-FT	[CLS] aggregation	0.6247 ± 0.0015	—
Dual-Tower Transformer†	mean pooling	0.6171 ± 0.0058	-0.76
DT-MLP†	no self-attention	0.5394 ± 0.0054	-8.53
DT-CA	cross-attention	0.6108 ± 0.0099	-1.39
DT-AS	soft-normalisation	0.6096 ± 0.0065	-1.52
DT-DC	demographic cross-attention	0.6192 ± 0.0067	-0.56

CROSS-SEED STABILITY

0.0015 AUPRC SD
≈4× lower cross-seed SD than the prior baseline (observed; prior run, not retrained).
Worst seed AUPRC: **0.6233**

Mean-pool variant: −0.76 pp.
DT-MLP† (no self-attention): −8.53 pp.
† From the prior dual-tower work (same cohort, not retrained); observed differences, not controlled ablations.

07 Runtime Safeguard

stress test

0% crashes · safeguard ON
100% batches failed · safeguard OFF

TESTED INPUT PERTURBATIONS
1% of test rows perturbed after scaling · all on numerical columns

- `Age = 200`
- `NaN (SaO₂)`
- `Glucose ×10`
- `BUN = -1`

Safeguard ON completed all five tested conditions with **0% crashes**; AUPRC stayed within **±0.0012** of the clean baseline.

OFF fails even on clean input because the deployed scaler standardises categorical indices, making the runtime clamp necessary.

08 MIMIC-IV Cross-Database Transfer

Table 3

No evaluated variant exceeded the eICU-only DT-FT reference.

Variant · training	AUPRC	Δ pp
DT-FT reference · eICU only	0.6247 ± 0.0015	0.00
DT-FT, combined · eICU + MIMIC-IV pooled	0.6188 ± 0.0080	-0.59
DT-FT, 2-stage · combined → eICU fine-tuning	0.6169 ± 0.0066	-0.78
DT-FT-MA · eICU only · architecture-matched control	0.6240 ± 0.0057	-0.07
DT-FT-MA, 2-stage · combined → eICU fine-tuning	0.6226 ± 0.0048	-0.21

POSSIBLE EXPLANATIONS — *hypotheses only; no isolating ablation was performed.*

- **Feature-schema mismatch:** six eICU clinical measurements are absent from MIMIC-IV.
- **Domain mismatch:** multi-center eICU vs. single-center MIMIC-IV.
- **Encoding mismatch:** gender, ethnicity, age and site_id codes differ between the two preprocessing pipelines (see GitHub).

09 Conclusion, Limitations & Future Work

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DualTower-FT is positioned as a **deployment-engineering contribution**: an end-to-end deep tabular model with a forward-pass runtime safeguard and documented evidence on the limits of naive cross-database transfer.
Calibration remains imperfect, and prospective clinical validation is required before deployment.

LIMITATIONS

- **Retrospective study** with a single binary outcome: in-hospital mortality.
- **One time-point** feature vector per patient; longitudinal trajectories are not modelled.
- Generalisation to **unseen hospitals** was not directly tested; the split was by ICU stay rather than leave-hospital-out.
- Stress testing used a single **1% corruption rate**, and all four conditions targeted numerical columns.
- **Comparison** omitted CatBoost, LightGBM, and newer tabular architectures.
- Baseline comparisons were not fully controlled because prior models were **not retrained** with this codebase.
- **Interpretability and uncertainty** estimation were outside the present scope.

FUTURE WORK

- Longitudinal trajectory modelling
- Clinical-note integration
- Domain adaptation for cross-database transfer
- Deployment-time recalibration